Midterm Report: Environments for testing Methods for Lie Detection in LLMs

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Abstract

Can deception detectors help agents make better decisions in deception-related environments? Your abstract goes here. Describe your research question, why it matters for AI safety, your approach, and preliminary findings (100-200 words).

Keywords alignment research · machine learning · AI safety

1 Introduction

Understanding how large language models lie is becoming more important as their capabilities increase. One way of evaluating large langauge models has been to use what is called a black-box evaluator, a different large-langauge model that checks the output for falsehoods. However, such models have limitations and can be computationally expensive. White-box detectors complement this approach by applying simpler statistical models to the internal datastreams of LLMs. As LLMs are increasingly used as agents in helping with research, booking-systems and coding, there is a growing need for evaluating these methods in settings where they are most likely to be used. Previous research has found th lie-detection methods often fail to generalize to new datasets or model families. Defining what a lie is is also methodologically fraught with risk. As such, creating correct labels for datasets can often result in noisy evaluations. To address this problem in evaluating lie detection, we have tested two previously untested environments for evaluating lie detection methods in large language models over X many different model families, sizes and probe architectures.

Our main contributions are as follows:

- first
- second
- third

2 Methodology

Describe your research approach and methods.

3 Preliminary Results

- We evaluated Carrot-parsnip as a simple social deduction game to test lie detection methods in
- We found that strong auditors (Opus 5.5) can benefit from weak probes in this setting
- We also found that if the probe is too weak, strong auditors can be hurt by weak probes

4 Challenges & Open Questions

- How do we differentiate between a weak probe and a broken environment?
- How do we make sure that our experiments and environment design is robust to different kinds of probes?

5 Next Steps

Describe your planned next steps for the remainder of the program. What will you prioritize?

References